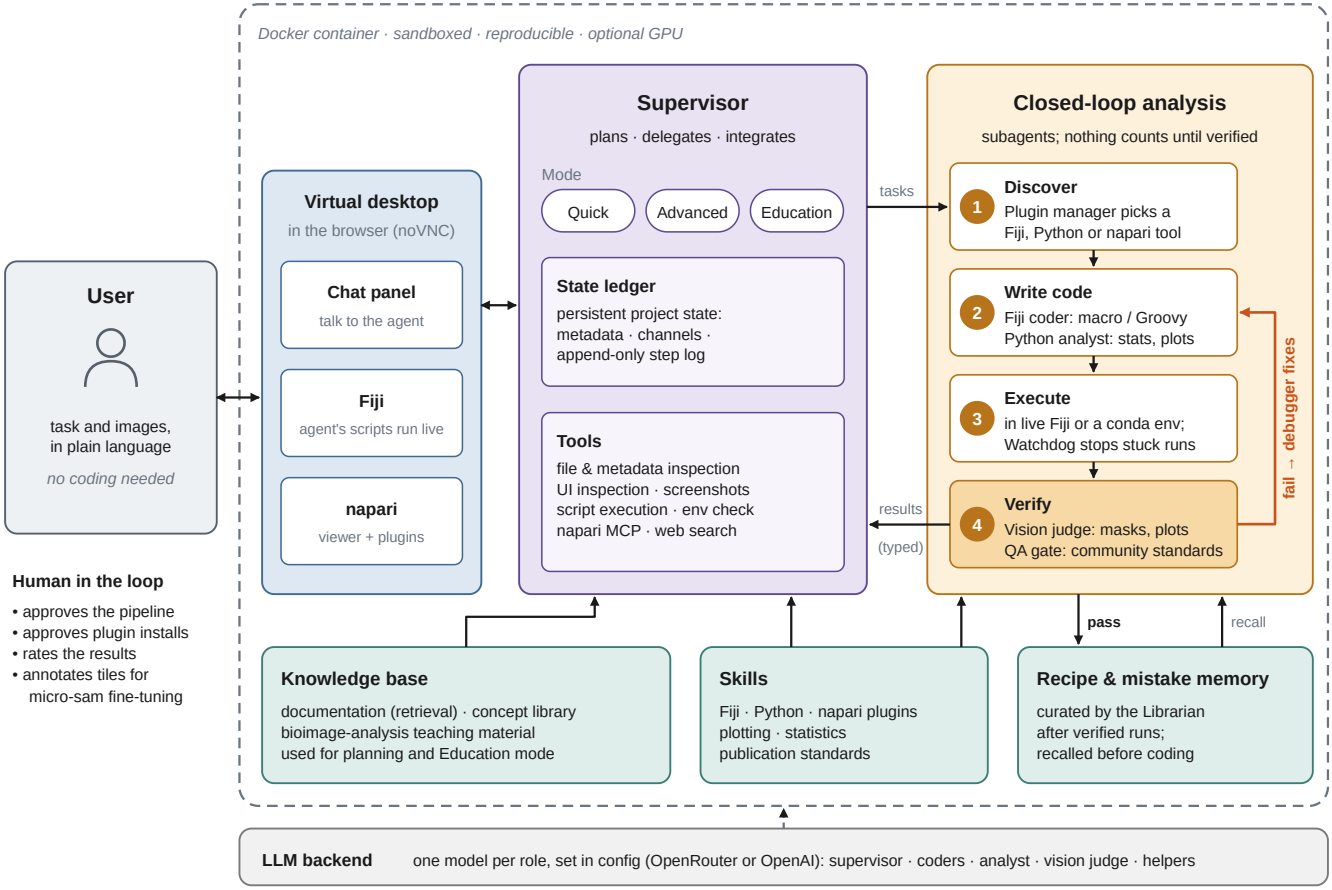

Docker container · sandboxed · reproducible · optional GPU
User
task and images, in plain language
no coding needed
Human in the loop
• approves the pipeline
• approves plugin installs
• rates the results
• annotates tiles for micro-sam fine-tuning
Virtual desktop
in the browser (noVNC)
Chat panel
talk to the agent
Fiji
agent's scripts run live
napari
viewer + plugins
Supervisor
plans · delegates · integrates
Mode
Quick
Advanced
Education
State ledger
persistent project state:
metadata · channels ·
append-only step log
Tools
file & metadata inspection
UI inspection · screenshots
script execution · env check
napari MCP · web search
tasks
results
(typed)
Closed-loop analysis
subagents; nothing counts until verified
1 Discover
Plugin manager picks a Fiji, Python or napari tool
2 Write code
Fiji coder: macro / Groovy
Python analyst: stats, plots
3 Execute
in live Fiji or a conda env;
Watchdog stops stuck runs
4 Verify
Vision judge: masks, plots
QA gate: community standards
fail → debugger fixes
pass
recall
Knowledge base
documentation (retrieval) · concept library
bioimage-analysis teaching material
used for planning and Education mode
Skills
Fiji · Python · napari plugins
plotting · statistics
publication standards
Recipe & mistake memory
curated by the Librarian
after verified runs;
recalled before coding
LLM backend
one model per role, set in config (OpenRouter or OpenAI): supervisor · coders · analyst · vision judge · helpers